03 Filling gaps: a FLOS atlas on non-adult bones with MAPOD4D®

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The objective of this project is to create a free/libre and open source resource for the anthropological study of non-adult bones, through the use of MAPOD4D®, a Multiverse of metaverses specifically developed for cultural heritage enhancement (Taglioretti et al. 2021), that will allow the user to navigate in a 3D world to view digital twins of real bones, obtained with the structured-light scanner EinScan-SP V1 (using the software ExScan S 3.1.4.3). Each metaplanet in the metaverse will pertain to one bone of the human body, and will include different examples of it in different growth stages, through both 3D models and digital photos. This metaverse will serve as a comprehensive atlas on non-adult bones, bridging the gap in the resources available today, which often fail to mention non-adult bones, or include pictures of only a few growth stages, which is extremely reductive, as a non-adult bone's morphology can change a lot in a matter of months, even less in the fetal stage.

One challenge of this project is the fact that a scanning protocol for human remains has not been developed yet, but the research on it is still ongoing. Different protocols need to be developed for different kinds of bones, depending on their size and shape, and the size of non-adult bones is sure to pose a problem. (Orsi et al. 2024a, Orsi et al. 2024b) At the moment, the metaverse includes two metaplanets, one that serves as an introduction to the process of anthropological studies, which can be used for educational purposes, while the second focuses on the 2nd metacarpal, selected for its size and shape. The bones studied and scanned for this metaverse come from the archaeological site of Treviglio. (Galante 2018) One future perspective to be considered is the use of multisensor acquisition, instead of only structured light, to potentially implement to the protocols.